

Adrian Alemany

Programmer | IA & Data Engineer | Google Cloud Solutions

ABOUT ME:

Data Scientist & AI Engineer with technical training in Google Cloud, Data Engineering, and Generative Artificial Intelligence. Hands-on experience in developing End-to-End architectures: from data ingestion and pipelines (dbt, Terraform, GCP) to the deployment of predictive models and RAG agents. Passionate about technical innovation and continuous upskilling within the Open Source ecosystem.

HARDSKILLS

Languages & Frameworks

- Python
- SQL
- JavaScript,
- React
- TypeScript.
- UV

IA

- LLMs
- RAG
- Agentes Autónomos(LangChain)
- Scikit-Learn
- XGBoost
- PyTorch
- TensorFlow
- MLOps(MLFlow, Grafana)

DATA

- Google Cloud Platform (Vertex AI, BigQuery, Dataflow, Pub/Sub)
- Apache Beam
- Kafka
- dbt
- Terraform
- Docker
- Web Scraping
- Ingesta con APIs

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CERTIFICATES

Profesional Machine Learning Engineer Cloud

Google

AI-300: Machine Learning Operations Engineer Associate

Azure

FEATURED TECHNICAL PROJECTS

Maritime Fuel Consumption Predictive System — Baleària Case Study

Stack: **Python**, GCP (**Pub/Sub**, **Dataflow**, **BigQuery**, **Vertex AI**), **DBT**, **Terraform**, **CI/CD** (**GitHub Actions**).

- Designed an **end-to-end** streaming and batch data **pipeline** utilizing Pub/Sub, Dataflow, and analytical transformations with DBT and BigQuery.
- Trained and deployed Machine Learning models on Vertex AI **Endpoints** to predict optimal fuel consumption per maritime route.
- Implemented Infrastructure as Code (IaC) via Terraform and ensured continuous deployment using CI/CD pipelines in GitHub Actions.
- Developed an interactive dashboard for real-time monitoring of predictions and vessel telemetry.

Intelligent Job Platform with Autonomous Agents

Stack: Python, **Google ADK**, **MCP**, **A2A React**, **TypeScript**, **Gemini API**, **Speech-to-Text**, **FastAPI**, **App Engine**.

- Built a multi-agent backend architecture (sequential and parallel) with Langchain for automated profile and job offer analysis, filtering, and matching.
- Integrated **MCPs** into the **orchestrator agent** to utilize tools such as the Speech-to-Text API and Gemini API for real-time interactive voice interviews with an **LLM**.
- Developed a reactive and typed interface in React + TypeScript, connected to **low-latency** backend microservices, and planned App Engine integration for future project updates.

Tourist Occupancy Predictive Engine (Airbnb Dataset)

Stack: Python, **XGBoost**, Vertex AI, **REST API**, DBT, Terraform.

- Developed a predictive model using XGBoost to estimate occupancy rates, ingesting and cross-referencing weather and real estate market data via external APIs.
- Orchestrated modular data preprocessing with DBT and provisioned infrastructure on Google Cloud with Terraform.
- Deployed the model for online inference on a managed Vertex AI endpoint.

CivicPulse AI — Multi-Agent Autonomous Governance Platform

Stack: Python, FastAPI, LangGraph, **PostgreSQL**, **Docker**, GCP (**Cloud Run**), **ChromaDB**, **Streamlit**.

- Designed and implemented an autonomous, distributed multi-agent system capable of simulating decision-making based on real-time data streams and algorithmic governance.
- Orchestrated asynchronous communication and consensus resolution among specialized agents using LangGraph, ChromaDB (RAG for regulations), and external public data APIs.
- Deployed the containerized architecture with Docker on Google Cloud Platform, optimizing token consumption and computing costs by 70% using Open Source models and local inference.
- Developed an interactive real-time dashboard and a REST API in FastAPI to monitor the system's status, reasoning traces, and automated publications.

HACKATHONS & COMPETITIONS

Kaggle Predictive Machine Learning Challenge

- Performed exploratory data analysis (**EDA**), feature engineering (**encoding**, **normalization**, **PCA**, **MICE**, etc.), and comparative experimentation between traditional models (**Scikit-Learn**) and Deep Learning architectures (**PyTorch** / **TensorFlow**).
- Optimized hyperparameters to maximize the evaluation metric up to **96% accuracy**, while cross-validating model performance using additional metrics (**Confusion Matrix**, **ROC AUC**, **PR AUC**, **Cross-Validation**).

Intelligent Price Comparator with RAG and Document Extraction (Hackathon)

Stack: Python, **Crawl4AI**, **Scrapy**, **Document AI (OCR)**, LangChain, **Vector DB**, FastAPI, Streamlit.

- Built an automated extraction and resilient scraping pipeline using Crawl4AI and Scrapy, processing catalogs and brochures via Document AI (OCR).
- Implemented a natural language querying system with **Retrieval-Augmented Generation (RAG)** to compare prices and suggest optimal purchasing options.
- Created an MVP in record time.

FORMATION

Master de IA, DATA & Google Cloud

EDEM

Bootcamp de creación de paginas web

EDEM